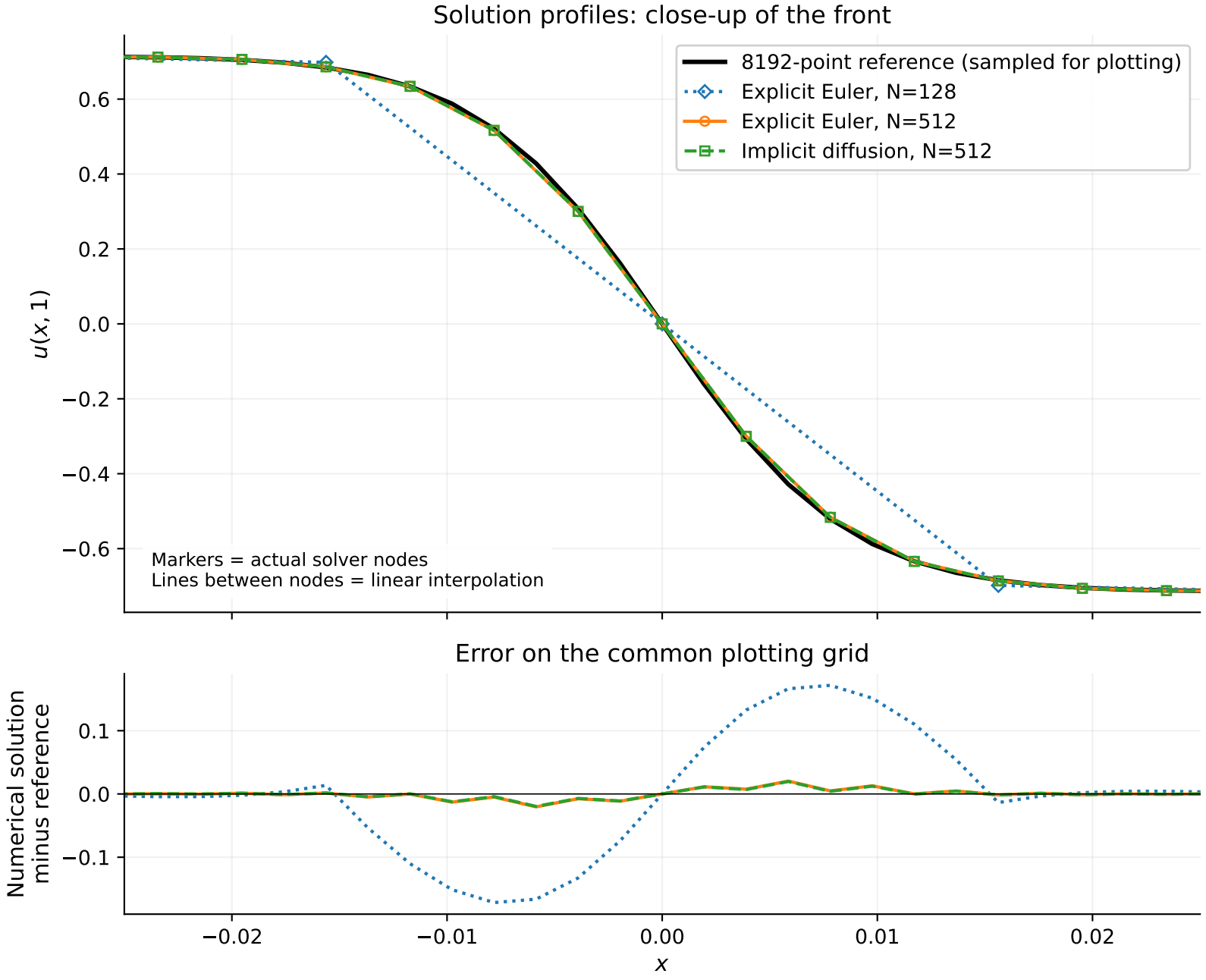
Solution profiles: close-up of the front
8192-point reference (sampled for plotting)
Explicit Euler, N=128
Explicit Euler, N=512
Implicit diffusion, N=512
u(x, 1)
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
Markers = actual solver nodes
Lines between nodes = linear interpolation
Error on the common plotting grid
Numerical solution
minus reference
0.1
0.0
−0.1
−0.02
−0.01
0.00
0.01
0.02
x